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Discovering Household Energy Consumption Regimes with Gaussian Hidden Markov Models

Temporal structure, probabilistic inference and comparison with
Gaussian Mixtures

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Motivation

Household electricity consumption is not just a collection of independent measurements.

It often follows latent regimes:

- ▶ baseline consumption;
- ▶ moderate domestic activity;
- ▶ kitchen activity;
- ▶ laundry activity;
- ▶ high-consumption periods.

The goal is to infer these regimes from observed energy measurements using probabilistic latent-variable models.



Research Question

Main question

Does modelling temporal dependence produce better and more coherent household energy-consumption regimes than an independent Gaussian mixture model?

Compared models

- ▶ Gaussian Mixture Model: independent latent assignments.
- ▶ Gaussian Hidden Markov Model: temporally dependent latent states.

Evaluation criteria

- ▶ held-out average log-likelihood;
- ▶ interpretability of inferred regimes;
- ▶ temporal persistence of state assignments.

Dataset

Dataset: UCI Individual Household Electric Power Consumption.

- ▶ Minute-level measurements.
- ▶ Period: December 2006 – November 2010.
- ▶ More than two million observations.
- ▶ Missing measurements: about 1.25% of rows.

Selected variables

- ▶ Global active power.
- ▶ Sub-metering 1: kitchen.
- ▶ Sub-metering 2: laundry.
- ▶ Sub-metering 3: water heater / air conditioning.



Preprocessing Pipeline

- 1 Combine date and time into a single timestamp.
- 2 Aggregate minute-level measurements into hourly energy.
- 3 Keep only complete hours with 60 valid minutes.
- 4 Detect continuous temporal sequences.
- 5 Split chronologically into train, validation and test.
- 6 Standardize features using training-set statistics only.

Split	Hours	Complete hours	Sequences
Training	24212	24022	51
Validation	5188	5090	12
Test	5189	4973	8

Gaussian Mixture Model Baseline

The GMM assumes that each hourly observation is generated independently from one of K Gaussian components.

$$z_n \sim \text{Categorical}(\pi)$$

$$x_n \mid z_n = k \sim \mathcal{N}(\mu_k, \Sigma_k)$$

$$p(x_n) = \sum_{k=1}^K \pi_k \mathcal{N}(x_n \mid \mu_k, \Sigma_k)$$

- ▶ Each component represents a consumption regime.
- ▶ Diagonal covariance matrices were used.
- ▶ Limitation: no transitions between consecutive hours are modelled.
- ▶ Parameters are learned by EM using multiple random initializations.

Gaussian Hidden Markov Model

The HMM introduces a latent state sequence with Markovian dependence.

$$z_1 \rightarrow z_2 \rightarrow \cdots \rightarrow z_T$$

$$x_t \mid z_t = k \sim \mathcal{N}(\mu_k, \Sigma_k)$$

The joint distribution factorizes as:

$$p(z_{1:T}, x_{1:T}) = p(z_1) \prod_{t=2}^T p(z_t \mid z_{t-1}) \prod_{t=1}^T p(x_t \mid z_t)$$

- ▶ z_t : hidden household-energy regime.
- ▶ x_t : observed hourly energy vector.
- ▶ The transition matrix A is the key difference from the GMM.
- ▶ Gaussian emissions use diagonal covariance matrices.

Learning the HMM: Baum–Welch

The Gaussian HMM is trained with Baum–Welch, an EM algorithm.

$$\theta = \{\pi, A, \mu_1, \dots, \mu_K, \Sigma_1, \dots, \Sigma_K\}$$

E-step

Use Forward–Backward to compute:

$$\gamma_t(k) = p(z_t = k \mid x_{1:T}), \quad \xi_t(i, j) = p(z_t = i, z_{t+1} = j \mid x_{1:T})$$

M-step

Update initial probabilities, transition probabilities and Gaussian emission parameters.

Multiple random initializations were used for each K to reduce sensitivity to local optima.

Inference Algorithms

The main HMM inference algorithms were implemented explicitly in log-space.

- ▶ **Forward algorithm**

$$\alpha_t(j) = p(x_{1:t}, z_t = j)$$

Computes the likelihood of the observed sequence.

- ▶ **Forward–Backward smoothing**

$$\gamma_t(k) = p(z_t = k \mid x_{1:T})$$

Computes posterior state probabilities.

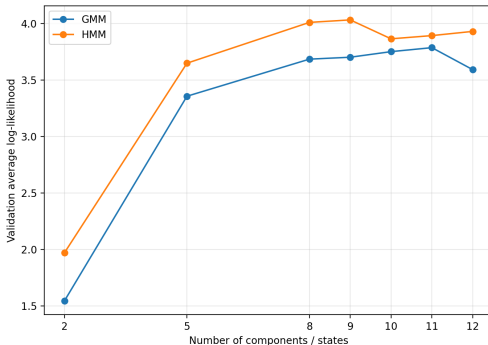
- ▶ **Viterbi algorithm**

$$z_{1:T}^* = \arg \max_{z_{1:T}} p(z_{1:T} \mid x_{1:T})$$

Finds the most likely hidden-state path.

Model Selection

Models were compared using validation average log-likelihood.



For every $K \in \{2, 5, 8, 9, 10, 11, 12\}$, the HMM achieves a higher validation average log-likelihood than the GMM. The test set was not used for model selection.

Model	Selected K	Validation LL
GMM	11	3.7876
HMM	9	4.0333

GMM Components

The selected GMM uses $K = 11$ components. The table shows representative consumption profiles.

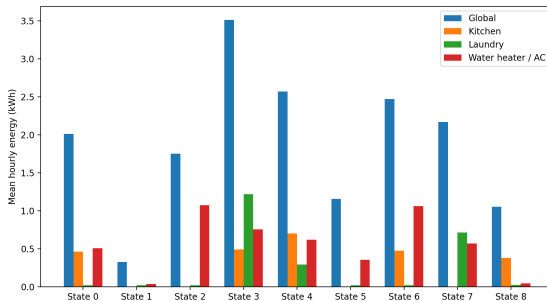
Instead of a small number of broad regimes, the GMM splits consumption into more specific independent components.

Component	Interpretation	Weight
3	Low / baseline consumption	42.23%
0	Moderate + water heater/AC	21.15%
2	Strong water heater/AC	9.35%
9	Kitchen activity	5.09%
7	Laundry + water heater/AC	5.13%
1	Very high / multiple appliances	1.67%

- ▶ The largest component is the low-consumption baseline.
- ▶ Additional components separate appliance-specific and high-load consumption patterns.
- ▶ Unlike the HMM, component assignments are independent across time.

HMM State Profiles

The selected HMM uses $K = 9$ hidden states.



State	Interpretation	State	Interpretation
0	Kitchen + moderate	5	Moderate
1	Baseline	6	Kitchen + water/AC
2	Water heater/AC	7	Laundry
3	Very high / multiple appliances	8	Kitchen
4	High mixed		

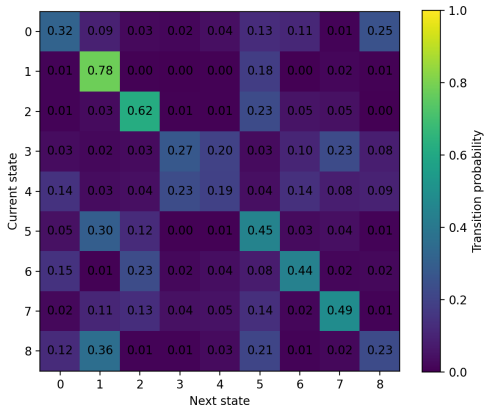
Interpretation

- ▶ Low-consumption periods
- ▶ Appliance-dominated activity
- ▶ Mixed high-consumption regimes

State indices are arbitrary.

Transition Matrix

The transition matrix describes how regimes evolve over time.



- ▶ Diagonal values measure state persistence.
- ▶ The baseline state is the most persistent one: $A_{11} \approx 0.78$.
- ▶ Kitchen activity (State 8) often transitions back to the baseline state (State 1), with $A_{81} \approx 0.36$.

Expected State Durations

For a standard HMM, state durations are geometrically distributed.

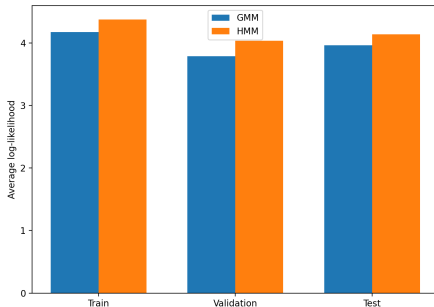
$$\mathbb{E}[D_k] = \frac{1}{1 - A_{kk}}$$

State	Interpretation	Expected duration
0	Kitchen + moderate use	1.46 h
1	Baseline	4.53 h
2	Water heater/AC	2.64 h
3	Very high / multiple appliances	1.38 h
4	High mixed activity	1.24 h
5	Moderate consumption	1.81 h
6	Kitchen + water heater/AC	1.79 h
7	Laundry-dominated	1.95 h
8	Kitchen activity	1.30 h

The baseline regime has the longest expected duration.

Average Log-Likelihood

The HMM assigns higher average log-likelihood on train, validation and test data.



Dataset	GMM	HMM	Difference
Training	4.1733	4.3781	+0.2048
Validation	3.7876	4.0333	+0.2457
Test	3.9623	4.1362	+0.1739

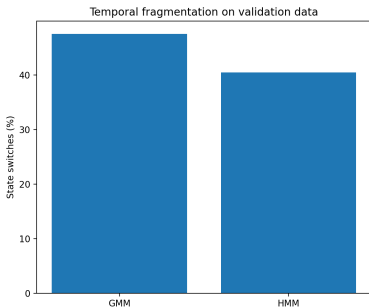
The HMM achieves higher average log-likelihood on all three datasets.

More importantly, the improvement remains on both validation and test data.

The test set was evaluated only after model selection.

Temporal Coherence

The selected HMM produces more persistent latent-state sequences than the selected GMM.



Metric	GMM	HMM
Switch rate	47.54%	40.47%
Mean run	2.10 h	2.46 h
Median run	1.00 h	2.00 h

This is a descriptive comparison of the selected models rather than a controlled same- K comparison, since they use different numbers of latent regimes.

Both models are evaluated on the same 5,078 consecutive-hour transitions.

Posterior Uncertainty

Forward-Backward gives posterior state probabilities:

$$\gamma_t(k) = p(z_t = k \mid x_{1:T})$$

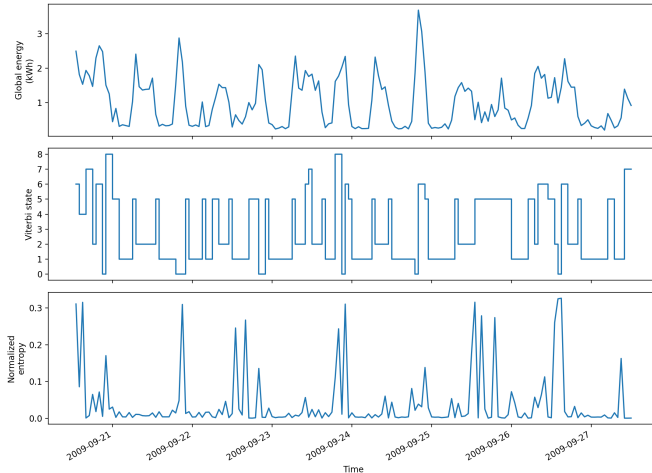
Posterior uncertainty was measured with normalized entropy:

$$H_t = -\frac{1}{\log K} \sum_k \gamma_t(k) \log \gamma_t(k)$$

Quantity	Value
Mean normalized entropy	0.0352
Median normalized entropy	0.0060
Mean maximum posterior probability	0.9723
Posterior confidence < 0.80	5.01%

Low entropy indicates confident model assignments, not classification accuracy, since true latent regimes are unobserved.

Example of Inferred Regimes Over Time



- ▶ Example: first continuous validation sequence, up to 168 hours.
- ▶ Top: global energy; middle: Viterbi state; bottom: posterior uncertainty.
- ▶ Some entropy peaks coincide with regime changes.

Conclusions

Modelling temporal dependence provides a better probabilistic description of household energy-consumption regimes.

- ▶ The HMM achieves higher validation likelihood for every tested K .
- ▶ For the selected models:

Validation: $3.7876 \rightarrow 4.0333$, Test: $3.9623 \rightarrow 4.1362$.

- ▶ Both models recover interpretable consumption regimes.
- ▶ The selected HMM is less temporally fragmented (40.47% vs. 47.54% switches), although the models use different selected K .
- ▶ The baseline regime is the most persistent and posterior state uncertainty is low for most hours.